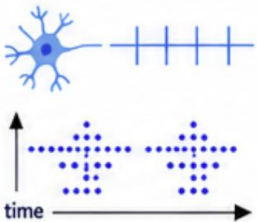


Neuron Encoding

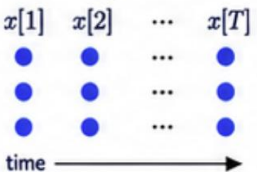
Image / Event Stream



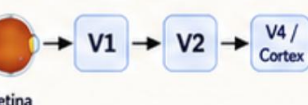
Neuron Encoding



Convert to T-step tokens



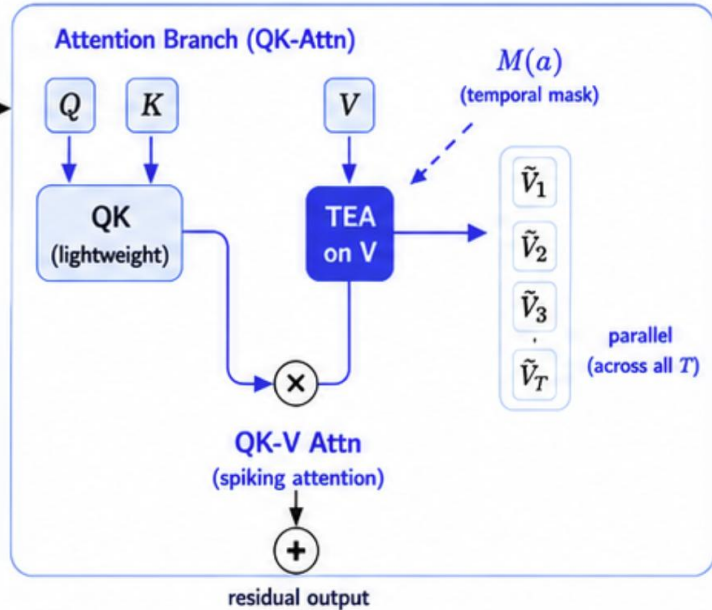
Brain-inspired pathways



→ Feedforward (fast)
→ Feedback (refinement)

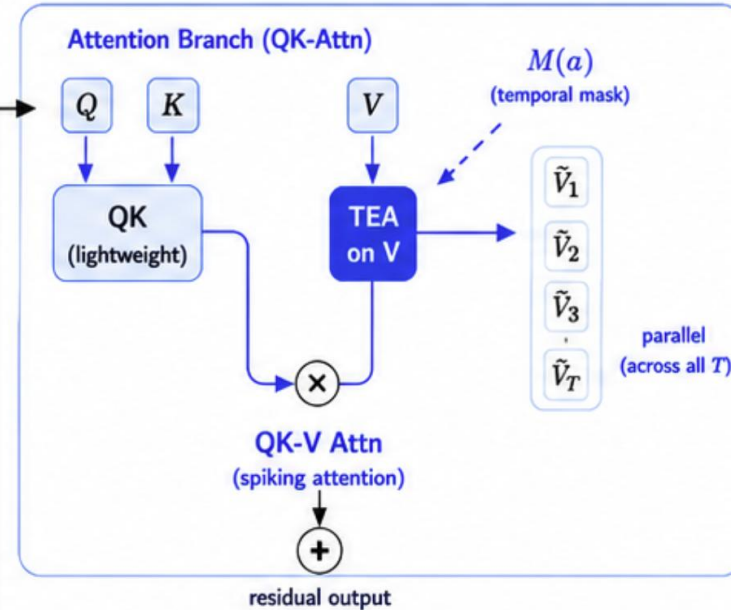
Shallow 50% Blocks (explicit temporal enhancement)

TEFormer Block 1



...

TEFormer Block $L/2$



Deep 50% Blocks (standard spiking processing)

Block $L/2 + 1$

QK-Attn / SSA

...

Block $L - 1$

QK-Attn / SSA

Block L

QK-Attn / SSA

no explicit temporal enhancement in deep layers; use QK-Attn for efficiency or SSA for finer modeling

Output

